



Computer Networks Lab

Fall 2025

Week 12

VLSM

Introduction:

VLSM is a Variable Length Subnet Mask in which the subnet design uses more than one mask in the same network which means more than one mask is used for different subnets of a single class A, B, C, or a network. It is used to improve the usability of subnets as they can be of variable size. It is also defined as the process of subnetting a subnet.

How to do Subnetting with VLSM?

Step 1: Determine Host Requirements

Identify the number of hosts needed in each LAN and WAN:

- **HQ LAN** – 50 hosts
- **BRANCH 1** – 30 hosts
- **BRANCH 2** – 20 hosts
- **WAN 1 (HQ to BRANCH 1)** – 2 hosts
- **WAN 2 (BRANCH 1 to BRANCH 2)** – 2 hosts

Total: 104 hosts

Step 2: Choose IP Subnet Class

Based on the host count, a Class C subnet (192.168.10.0) will suffice.

Step 3: Host Bits per Subnet

For the HQ LAN (50 hosts), 6 host bits will fulfill the requirement.

Step 4: Calculate Subnet Mask

The subnet mask for HQ LAN is /26 or 255.255.255.192.

Step 5: Determine Increment

With 6 host bits, the increment is 64.

Step 6: Network, Broadcast, and Usable IP Range

- **Network Address:** 192.168.10.0
- **Broadcast Address:** 192.168.10.63
- **Usable IP Addresses:** 192.168.10.1 to 192.168.10.62

Implementing VLSM

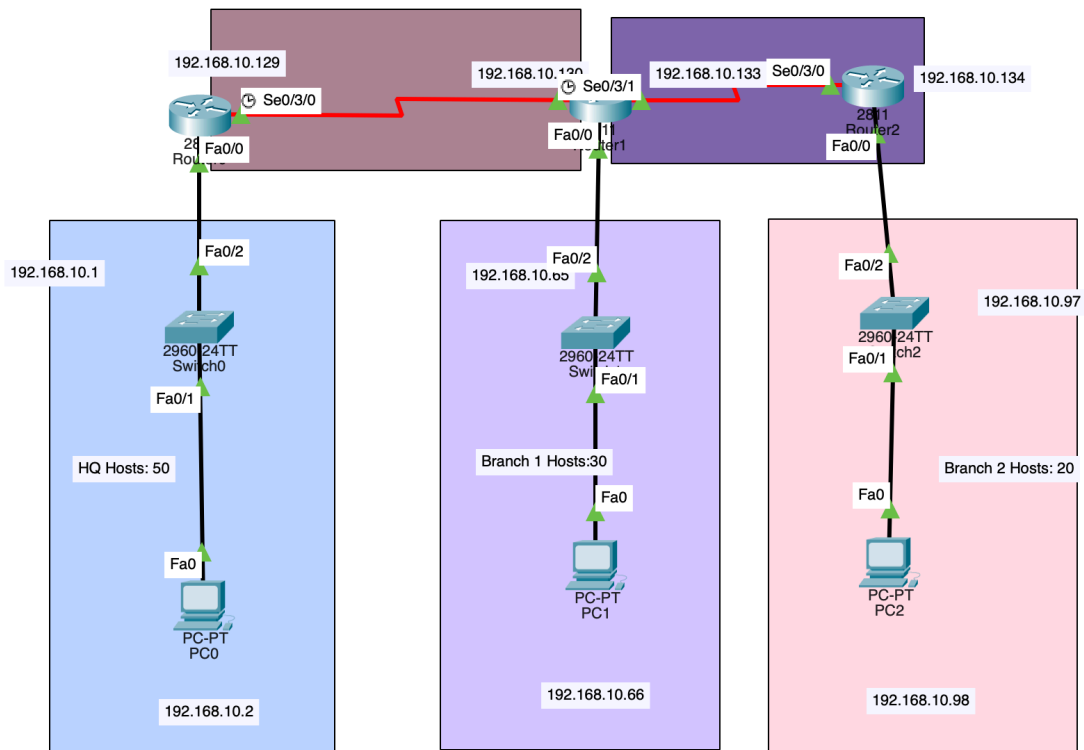
Subnetting:

- **BRANCH 1 LAN**
 - **Hosts:** 30
 - **Subnet Mask:** /27
 - **Network Address:** 192.168.10.64
 - **Broadcast Address:** 192.168.10.95
 - **Usable IP Range:** 192.168.10.65 to 192.168.10.94
- **BRANCH 2 LAN**
 - **Hosts:** 20
 - **Subnet Mask:** /27
 - **Network Address:** 192.168.10.96
 - **Broadcast Address:** 192.168.10.127
 - **Usable IP Range:** 192.168.10.97 to 192.168.10.126
- **WAN 1, WAN 2**
 - **Hosts:** 2 each
 - **Subnet Mask:** /30
 - **Usable IP Range:** Address ranges for each WAN

By following these steps, you can efficiently allocate IP addresses across various subnets within your network.

VLSM in CISCO PACKET TRACER

Make Subnets of the Network: 192.168.10.0/24 according to the required number of hosts



Steps to create VLSM in CISCO Packet Tracer

Step 1: First, open the cisco packet tracer desktop and select the devices given below:

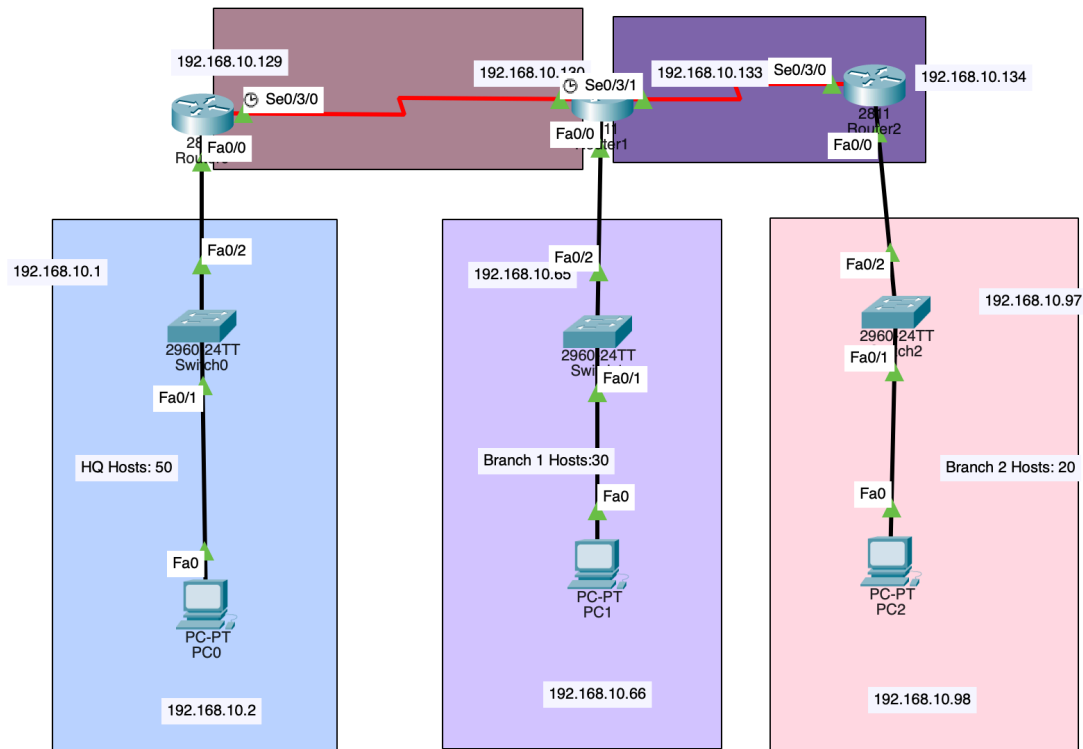
S.NO	Device	Model-Name	Qty
1.	pc	pc	3
2.	switch	PT-Switch	3
3.	router	PT-Router	3

IP Addressing Table for PCs

S.NO	Device	IPv4 Address	Subnet-Mask	Default-Gateway
1.	pc0	192.168.10.2	255.255.255.192	192.168.10.1
2.	pc1	192.168.10.66	255.255.255.224	192.168.10.65
3.	pc2	192.168.10.98	255.255.255.224	192.168.10.97

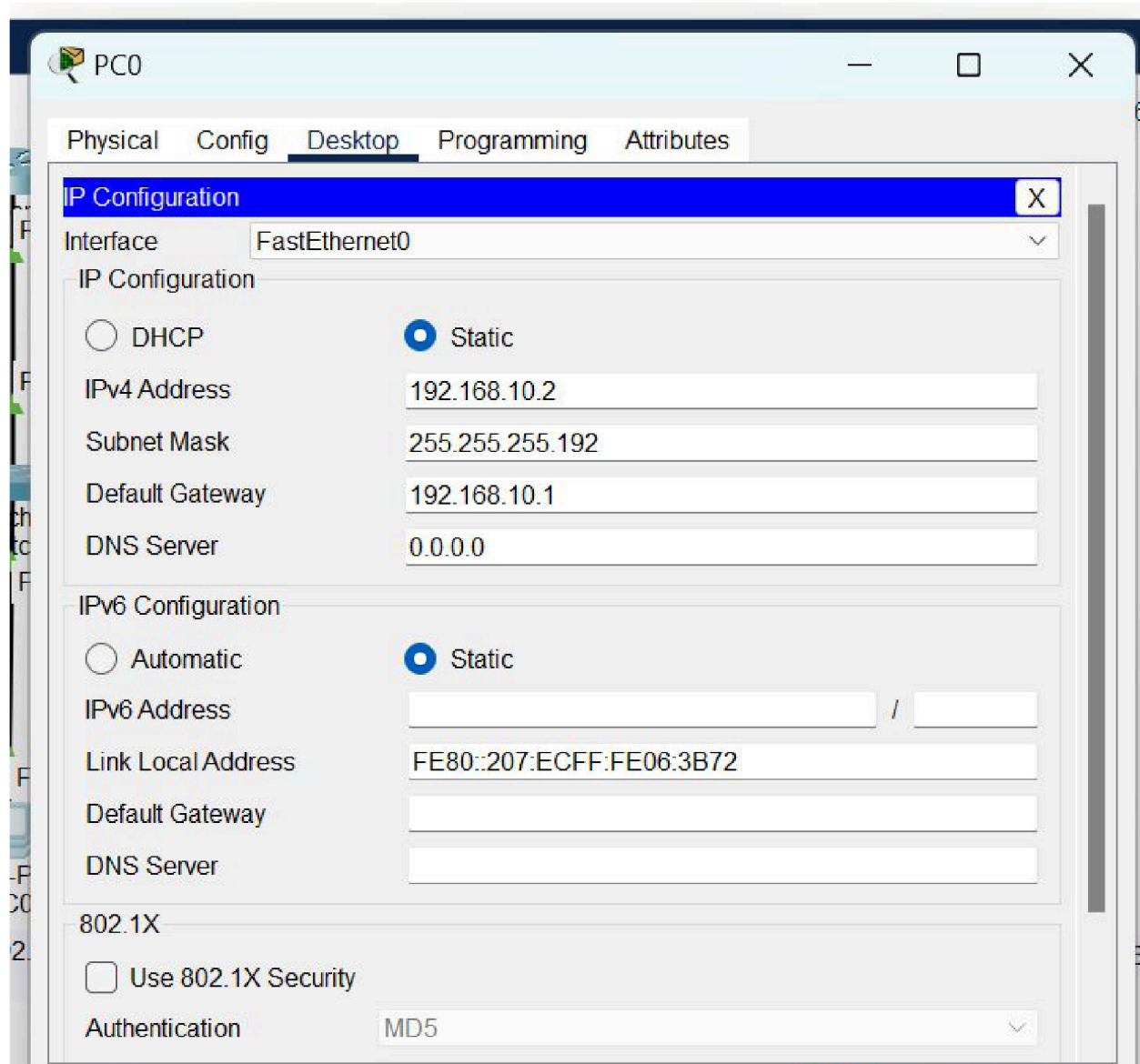
- Then, create a network topology as shown below the image.
- Use an automatic connecting cable to connect the devices with others.

Make Subnets of the Network: 192.168.10.0/24 according to the required number of hosts



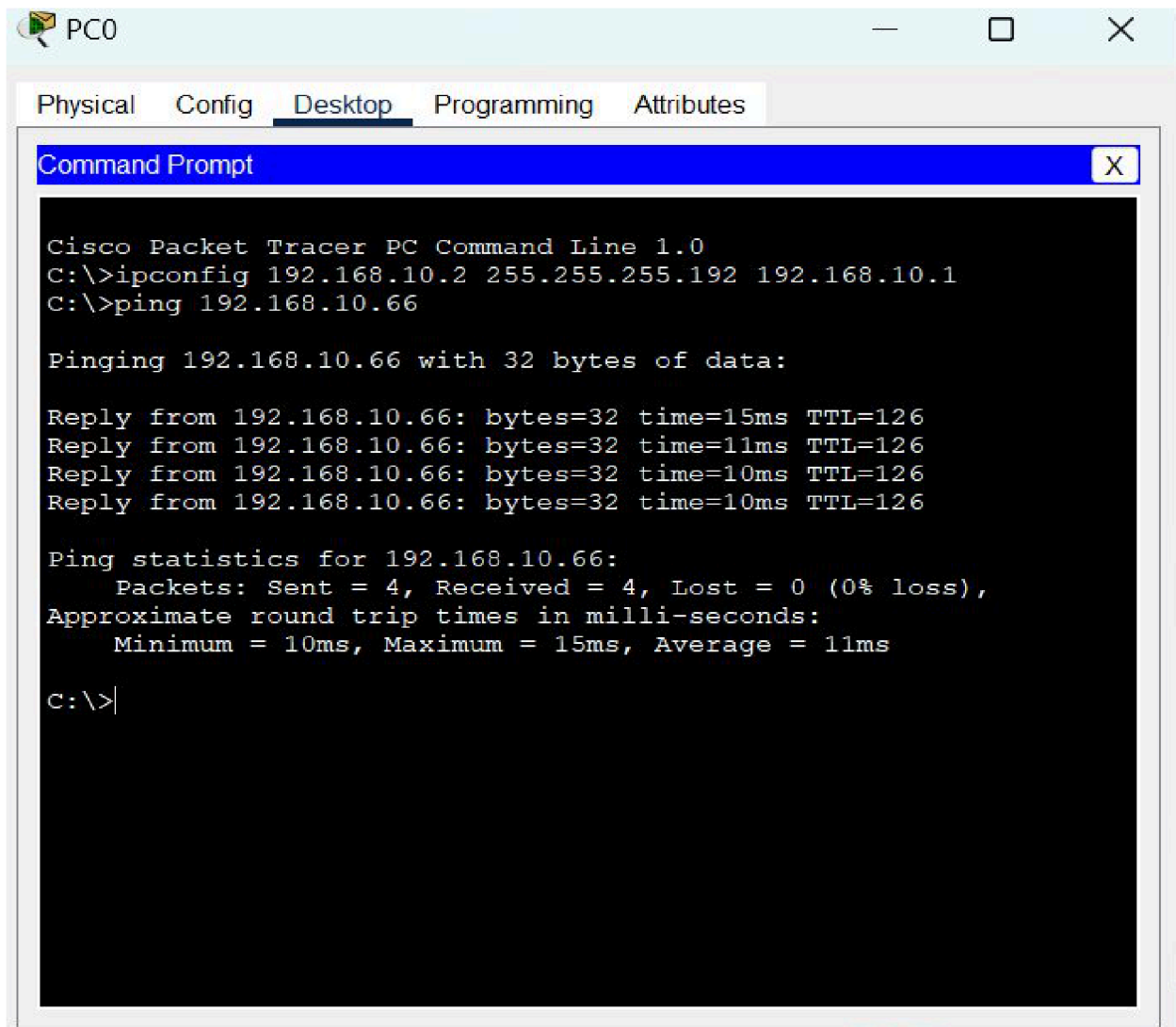
Step 2: Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

- To assign an IP address in PC0, click on PC0.
- Then, go to desktop and then IP configuration and there you will IPv4 configuration.
- Fill IPv4 address and subnet mask.



- Assigning an IP address using the ipconfig command, or we can also assign an IP address with the help of a command.
- Go to the command terminal of the PC.
- Then, type ipconfig <IPv4 address> <subnet mask> <default gateway>

Example: `ipconfig 192.168.10.2 255.255.255.192 192.168.10.1`



- Repeat the same procedure with other PCs to configure them thoroughly.

Step 3: Configure router with IP address and subnet mask.

S.N O	Device	Interface	IPv4 Address	Subnet mask
1.	router0	FastEthernet0/ 0	192.168.10.1	255.255.255.192
		Serial2/0	192.168.10.129	255.255.255.252
2.	router2	FastEthernet0/ 0	192.168.10.65	255.255.255.224
		Serial2/0	192.168.10.130	255.255.255.252
		Serial3/0	192.168.10.133	255.255.255.252
3.	router3	FastEthernet0/ 0	192.168.10.97	255.255.255.224
		Serial2/0	192.168.10.134	255.255.255.252

- To assign an IP address in router0, click on router0.
- Then, go to config and then Interfaces.
- Now, configure the IP address in FastEthernet and serial ports according to IP addressing Table.

- Fill IPv4 address and subnet mask.

The screenshot shows the configuration window for Router0. The 'Config' tab is selected, and the 'FastEthernet0/0' interface is chosen from the left-hand menu. The interface settings are displayed on the right, including Port Status (On), Bandwidth (100 Mbps), Duplex (Full Duplex), MAC Address (0060.2FBE.389E), IP Configuration (IPv4 Address: 192.168.10.1, Subnet Mask: 255.255.255.192), and Tx Ring Limit (10). Below the configuration panel, the 'Equivalent IOS Commands' are listed in a terminal-style box.

Router0

Physical **Config** CLI Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

INTERFACE

FastEthernet0/0

FastEthernet1/0

Serial2/0

Serial3/0

FastEthernet4/0

FastEthernet5/0

FastEthernet0/0

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0060.2FBE.389E

IP Configuration

IPv4 Address 192.168.10.1

Subnet Mask 255.255.255.192

Tx Ring Limit 10

Equivalent IOS Commands

```
Router(config-router)#end
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#interface FastEthernet0/0
Router(config-if)#
```

- Repeat the same procedure with other routers to configure them thoroughly.

Step 4: After configuring all of the devices we need to assign the routes to the routers.

To assign static routes to the particular router:

- First, click on router0 then Go to CLI.
- then type the commands and IP information given below.

CLI command: `ip route <network id> <subnet mask> <next hop>`

Static Routes for Router0 are given below:

```
Router(config)#ip route 192.168.10.64 255.255.255.224  
192.168.10.130
```

```
Router(config)#ip route 192.168.10.132 255.255.255.252  
192.168.10.130
```

```
Router(config)#ip route 192.168.10.96 255.255.255.224  
192.168.10.130
```

Static Routes for Router1 are given below:

```
Router(config)#ip route 192.168.10.96 255.255.255.224  
192.168.10.134
```

```
Router(config)#ip route 192.168.10.0 255.255.255.192  
192.168.10.129
```

Static Routes for Router2 are given below:

```
Router(config)#ip route 192.168.10.64 255.255.255.224  
192.168.10.133
```

```
Router(config)#ip route 192.168.10.128 255.255.255.252  
192.168.10.133
```

```
Router(config)#ip route 192.168.10.0 255.255.255.192  
192.168.10.133
```

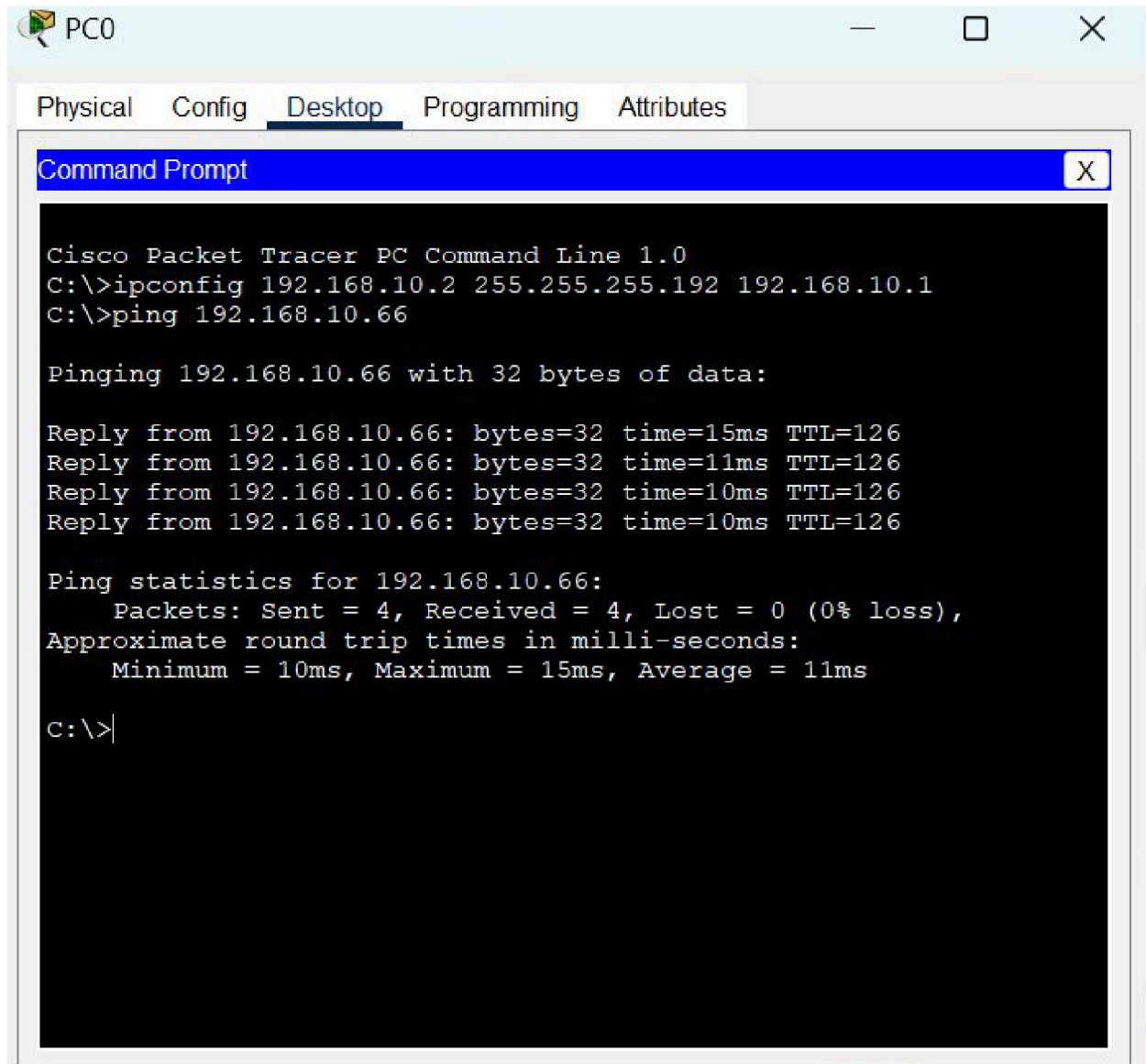
Step 5: Verifying the network by pinging the IP address of any PC.

we will use the ping command to do so.

- First, click on PC0 then Go to the command prompt.
- Then type ping <IP address of targeted node>.

- As we can see in the below image we are getting replies which means the connection is working.

Example : ping 192.168.10.66



- A simulation of the experiment is given below we are sending PDU from PC0 to PC1 and PC1 to PC2